43 questions | **CLO** 4-A | **Final** 29/12/2026

Q Exam-style questions: Sections 4-3, 4-4 and 4-5

أسئلة بمعايير اختبارات جامعة السلطان قابوس — الاحتمالات 4-3 و 4-4 و 4-5

Covers sections 4-3 (independent and dependent events, multiplication rules 1 and 2, conditional probability, independence tests, two-way tables, tree diagrams, at least one), 4-4 (fundamental counting rule, factorial, permutation rules 1 and 2, combinations) and 4-5 (probability with counting rules). Every question is tagged with its section.

طريقة الاستخدام

حل الجزء D أولًا (20 دقيقة) لتعرف مستواك، ثم الأجزاء A و B و C، ثم صحّح من ملف الإجابات. حدّد دائمًا: مع إرجاع أم بدون؟ الترتيب مهم أم لا؟ واكتب القاعدة التي تستخدمها، فعليها درجات.

Part A · Multiple choice (16)

Give the letter and a one-line reason, as in a written exam.

A1 4-3

Two cards are drawn from a standard deck **without replacement**. P(both are aces) =

(a) 1/169 (b) 1/221

(c) 1/26 (d) 2/13

A2 4-3

P(A) = 0.4, P(B) = 0.5 and P(A ∩ B) = 0.2. Which statement is true?

(a) A and B are mutually exclusive (b) A and B are independent

(c) P(A | B) = 0.5 (d) P(A ∪ B) = 0.9

A3 4-3

P(A ∩ B) = 0.12 and P(A) = 0.30. Then P(B | A) =

(a) 0.036 (b) 0.40

(c) 0.25 (d) 0.18

A4 4-3

Two events with non-zero probabilities are mutually exclusive. They must be:

(a) independent (b) dependent

(c) complementary (d) equally likely

A5 4-3

A fair coin is tossed 5 times. P(at least one head) =

(a) 5/32 (b) 1/2

(c) 31/32 (d) 1/32

A6 4-3

A bag has 4 red and 6 blue balls. Two are drawn **with replacement**. P(red first, then blue) =

(a) 0.24 (b) 0.2667

(c) 0.48 (d) 0.10

A7 4-3

A and B are independent with P(A) = 0.3 and P(B) = 0.6. P(A ∪ B) =

(a) 0.90 (b) 0.18

(c) 0.72 (d) 0.82

A8 4-4

The value of 6!/4! is:

(a) 1.5 (b) 2

(c) 30 (d) 720

A9 4-4

How many 4-letter codes can be made from the 26 English letters if letters may repeat?

(a) 26 × 25 × 24 × 23 (b) 456,976

(c) 14,950 (d) 104

A10 4-4

From 10 club members, a president, a vice-president and a secretary are chosen. The number of ways is:

(a) 120 (b) 720

(c) 30 (d) 1000

A11 4-4

$_{10}C_7$ equals:

(a) $_{10}C_3 = 120$ (b) $_{10}P_7 = 604{,}800$

(c) 70 (d) 7!

A12 4-4

The number of distinct arrangements of the letters of LEVEL is:

(a) 120 (b) 60

(c) 30 (d) 20

A13 4-4

A student has 3 shirts, 4 trousers and 2 pairs of shoes. How many different outfits are possible?

(a) 9 (b) 24

(c) 12 (d) 6

A14 4-5

A committee of 3 is chosen at random from 5 men and 4 women. P(all three are women) =

(a) 4/84 = 1/21 ≈ 0.0476 (b) 4/9

(c) 1/3 (d) 1/126

A15 4-5

A lottery draws 3 different numbers from 1 to 10; order does not matter. The probability that one ticket wins is:

(a) 1/720 (b) 1/120

(c) 3/10 (d) 1/1000

A16 4-5

The letters of MATH are arranged at random. P(the arrangement starts with M) =

(a) 1/24 (b) 1/4

(c) 1/6 (d) 4/24

Part B · Short structured questions (16)

Typical 2–6 mark midterm items. Write S and E in set notation.

B1 4-3

[5 marks] 200 SQU students were asked whether they use the university gym:

	USES GYM	DOES NOT	TOTAL
Male	48	52	100
Female	32	68	100
Total	80	120	200

Find: (a) P(uses gym | male) (b) P(male | uses gym) (c) P(male and uses gym) (d) Are "male" and "uses gym" independent? Justify with numbers.

B2 4-3

[5 marks] A box holds 20 phone chargers, 5 of them defective. Two are taken **without replacement**. Find: (a) P(both defective) (b) P(exactly one defective) (c) P(at least one defective). (d) Check that (a), (b) and P(none defective) add to 1.

B3 4-3

[4 marks] 60% of students come to SQU by bus. **Of the bus users**, 10% arrive late; of the others, 4% arrive late. Draw a tree diagram, then find: (a) P(bus and late) (b) P(late) (c) P(not bus and on time).

B4 4-3

[4 marks] $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cup B) = 0.7$. Find $P(A \cap B)$, $P(A \mid B)$ and $P(B \mid A)$. Are A and B independent? Are they mutually exclusive?

B5 4-3

[3 marks] A device has 3 components that work independently; each fails with probability 0.05. Find P(all three work) and P(at least one fails).

B6 4-3

[2 marks] Find the mistake. Given $P(A \cap B) = 0.15$ and $P(B) = 0.5$, a student writes "$P(A \mid B) = 0.15/P(A)$".

B7 4-3

[3 marks] Two dice are rolled. A = "the first die is even" and B = "the sum is 7". Show whether A and B are independent.

B8 4-4

[3 marks] How many 4-digit PINs (digits 0–9) are possible if (a) repetition is allowed (b) repetition is not allowed (c) no repetition and the first digit cannot be 0?

B9 4-4

[4 marks] A club has 12 members, including Ahmed. In how many ways can (a) a president, vice-president and secretary be chosen (b) a committee of 3 be chosen (c) a committee of 3 that includes Ahmed be chosen (d) a committee of 3 that excludes Ahmed be chosen?

B10 4-4

[3 marks] How many distinct arrangements can be made from the letters of SULTANATE? How many of them begin with S?

B11 4-4

[3 marks] A committee of 5 is formed from 6 men and 5 women. In how many ways can it have (a) exactly 2 men (b) at least 4 women?

B12 4-4

[2 marks] Find n if ${}_nC_2 = 45$.

B13 4-5

[3 marks] A bag holds 6 red and 4 white balls. Four are drawn at random. Find P(exactly 2 red) and P(at least 1 white).

B14 4-5

[3 marks] The letters of BANANA are arranged at random. Find P(the arrangement begins and ends with A).

B15 4-5

[2 marks] A 3-digit code is formed with no repeated digits (0–9). Find P(all three digits are odd).

B16 4-3 4-5

[3 marks] 12 bulbs include 3 that are defective. Three are chosen at random. Find P(none defective) in two ways: with combinations, and with the multiplication rule. Then find P(at least one defective).

Part C · Long questions (5)

Midterm-style, multi-part. Marks in brackets.

C1 4-3 **[12 marks]**

400 adults in Muscat were asked whether they exercise regularly:

AGE GROUP	YES	NO	TOTAL
18–30	90	60	150
31–50	70	80	150
over 50	40	60	100
Total	200	200	400

1. P(exercises)
2. P(exercises | 18–30)
3. P(over 50 | does not exercise)
4. P(18–30 and exercises)
5. Are "18–30" and "exercises" independent? Justify.
6. Two adults are chosen without replacement. Find P(both are over 50).

C2 4-3 **[10 marks]**

Box A has 3 red and 2 blue balls; box B has 1 red and 4 blue. A die is rolled: 1 or 2 → a ball is drawn from box A; otherwise from box B.

1. Draw the tree diagram with all branch probabilities.
2. P(box A and red)
3. P(red)
4. Given that the ball is red, find the probability it came from box A.
5. Are "box A" and "red" independent?

C3 4-4 4-5 **[12 marks]**

A committee of 5 is chosen at random from 7 engineers and 5 doctors.

1. Find n(S).
2. P(exactly 3 engineers)
3. P(all members have the same profession)
4. P(at least one doctor)
5. P(a particular engineer, Salim, is on the committee)

C4 4-4 4-5 **[8 marks]**

SQU student IDs have 6 digits, and the first digit must be 1 or 2.

1. How many IDs are possible if repetition is allowed?
2. How many if no digit may repeat?
3. With repetition allowed, find P(a randomly issued ID ends in 0).
4. With repetition allowed, find P(all six digits are the same).

C5 4-3 4-4 **[8 marks]**

A multiple-choice quiz has 5 questions, each with 4 options. A student guesses every answer independently.

1. How many different answer sheets are possible?
2. P(all five correct)
3. P(none correct)
4. P(at least one correct)

Part D · 20-minute mini-quiz (16 marks)

Time yourself, closed book, before opening the answers.

D1 4-3

[2 marks] Define independent and dependent events, with one example each.

D2 4-3

[3 marks] 30% of a city's residents own a car; of the car owners, 45% also own a motorbike. Find P(car and motorbike). Is 0.45 a conditional probability?

D3 4-4

[3 marks] Evaluate ${}_8P_3$, ${}_8C_3$ and 7!/(5! 2!).

D4 4-4

[2 marks] In how many ways can 7 different books be arranged on a shelf if a particular book must be in the middle?

D5 4-5

[3 marks] 4 students are chosen at random from 8 males and 6 females. Find P(2 males and 2 females).

D6 4-3

[3 marks] P(A) = 0.6, P(B | A) = 0.5 and P(B | A^c) = 0.2. Find P(A ∩ B) and P(B).

Built to the STAT 2101 Fall 2026 course outline (CLO4-A: calculate probabilities using counting rules and basic probability principles). Sections 4-3 Multiplication Rules and Conditional Probability, 4-4 Counting Rules and 4-5 Probability and Counting Rules only. Every numeric answer was computed by code from the question data.